Reg. No.									

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

SEVENTH SEMESTER

END SEMESTER EXAMINATIONS, NOVEMBER/DECEMBER 2024

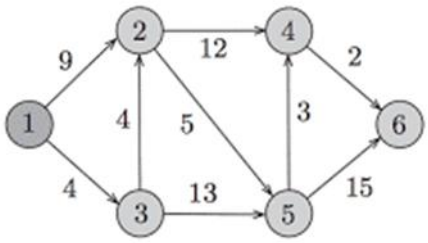
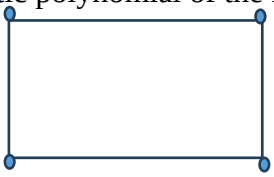
Graphs & Matrices [MAT 4054]

REVISED CREDIT SYSTEM

Time: 3 Hours **Date: 2 January 2025** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.

Q.NO	Questions	Marks	CO	BTL
1A.	Show that a simple graph with n vertices and k components cannot have more than $\frac{(n-k)(n-k+1)}{2}$ lines. Hence deduce that, if $q > \frac{1}{2}(p-1)(p-2)$, then G must be connected	4	1	4
1B.	Show that a graph is bipartite if and only if all its cycles are even	3	1	4
1C.	For any graph G, show that either G or its complement must be connected	3	1	4
2A.	Consider the following graph.  Use Dijkstra's algorithm to find the shortest path from 1 to all other vertices of the graph	4	2	3
2B.	For a non-trivial connected graph G with p points show that $\alpha_0 + \beta_0 = p$, where α_0 and $\beta_0$denote point covering number and point independence number respectively	3	3	4
2C.	Give example for each of the following graphs (i) an Eulerian graph (ii) an Hamiltonian graph that is not Eulerian (iii) neither Eulerian nor Hamiltonian	3	3	2
3A.	Obtain chromatic polynomial of the following graph 	4	3	3

	Use the recurrence relation f(G, t) = f(G + e, t) + f(G.e, t)			
3B.	For any graph G, prove that $k(G) \leq \lambda(G) \leq \delta(G)$, *where* $k(G), \lambda(G)$ *and* $\delta(G)$ respectively denote vertex connectivity, edge connectivity and minimum degree of the graph G	***3***	***3***	***4***
3C.	Let G be (p, q) graph. If G is tree then show that G is acyclic and that p = q+ 1	***3***	***2***	***4***
4A.	Let G be a graph with {0, 1, -1} incidence matrix Q(G). Then show that Q(G) is totally unimodular	***4***	***5***	***4***
4B.	If G is a connected graph on n vertices, let Q(G) denote the {0, 1, -1} incidence matrix of G. Show that rank of Q(G) = n-1	***3***	***5***	***4***
4C.	Let G be a connected graph with n vertices and M be the 0-1 incidence matrix of G. Then show that rank of M is n-1 if G is bipartite and n otherwise	***3***	***5***	***4***
5A.	For a plane map with p vertices, q edges and f faces show that p - q + f = 2	***4***	***3***	***4***
5B.	Let G be a bipartite graph with adjacency matrix A. If λ is an eigen value A with multiplicity k, then show that $-\lambda$ is also an eigen value of A with the same multiplicity	***3***	***4***	***4***
5C.	Let G be a simple graph with at least one edge. Let λ_1 be the largest eigen value of the Laplacaian matrix of G. Show that $\lambda_1 \geq \Delta(G) + 1$, where λ_1 denotes the largest eigen value of the Laplacian matrix and $\Delta(G)$ denotes the maximum degree of G	***3***	***4***	***3***